

Chinmay Raut

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EDUCATION

Indian Institute of Technology Madras

Chennai, India

Dual Degree Programme - B.Tech in Engineering Design & M.Tech in Biomedical Design

Jul 2018 – May 2023

- **CGPA: 8.91/10**
- Graduated in the **top 5% of the class**
- Relevant coursework: Applied Statistics, Mathematical Foundations of Data Science, Life Sciences, Human Anatomy, Physiology & Biomechanics, Medical Image Analysis, Signals and Systems, Digital Signal Processing, Design of Monitoring and Diagnostic Systems.

EXPERIENCE

Project Research Scientist - 1 — Health Informatics and Imaging Lab, IIT Madras | 🔍

Oct 2025 – Present

- Developed **PS-MAE** for an ICMR-funded breast cancer research. Designed its masked-pretraining objective using clinical evidence that peritumoral tissue contains metastasis-related signals. The encoder outperformed state-of-the-art methods under linear probing and on a class-imbalanced external cohort. First-author manuscript under review at IEEE JBHI.
- **JEPA-distilled USFMAE into a TinyViT** and added an auxiliary radiomics-prediction objective to capture information beyond the teacher embeddings. The 15× smaller encoder outperformed state-of-the-art classification models, retained performance on an external dataset, and performed best under linear probing, demonstrating inherently separable representations. Coordinated with clinicians to adapt the system for on-site deployment.
- Built an agentic system for the **MICCAI 2026 CHIMERA Challenge** that combined histopathology, clinical, and molecular evidence for prostate-cancer risk assessment. The system reported its confidence, influential variables, and policy trace, making each decision auditable.
- Evaluated breast-ultrasound segmentation across Indian, Polish, and Egyptian cohorts, identifying performance gaps caused by acquisition differences and population-linked morphology. Developed adaptive wavelet normalization to reduce this cross-cohort gap. Manuscript under review at BSPC.
- Mentored undergraduate and MS students, conducted tutorials on vision transformers, LLMs, Linux, and remote computing, and administered the lab's GPU infrastructure, including user access, storage, software environments, drivers, and GitHub repositories.

Product Analyst — Ather Energy, Bangalore, India | 🔍

Aug 2023 – Oct 2025

- Analysed **large-scale telemetric datasets** from 400,000+ connected electric scooters each having 100+ IoT sensors **transmitting 800 GB+ data daily**, extracting insights on riding style, charging patterns, smart features usage and vehicle utilisation across various demographic groups in India, Sri Lanka and Nepal.
- Built an ETL pipeline using **PySpark and Clickhouse SQL** for product analytics. Designed and monitored product metrics; conducted regular analysis on their movement to generate actionable product and business insights.
- Performed **population modelling** for ride assist features, correlating adoption and engagement of a feature to other ride metrics and behavioural patterns to build adaptive intelligent algorithms, which led to **12% improvement** in adoption.
- Developed a **LangChain-based LLM tool** to categorise user feedback from multiple channels, unifying qualitative data with quantitative metrics.

Undergraduate Researcher — iNSPIRE Lab, IITM, Chennai, India | 🔍

May 2021 – Dec 2021, Jul 2022 – May 2023

- Worked on developing tools and algorithms for planning needle trajectories for minimally invasive neurosurgeries.
- Implemented and trained a **UNet-based** network architecture using **Pytorch for tumour segmentation** using a publicly available dataset from the BraTS challenge.
- Utilised **FastsurferCNN for cortical parcellation and MRA data for blood vessel identification** to generate the environment for reinforcement learning.
- Developed a reinforcement learning framework called Flexible Needle Path Generation, which generated **15% safer and 10% smoother** curves compared to traditional path-finding algorithms like RRT*.
- Developed a virtual reality pipeline using **Slicer 3D software** and an Aruco marker as an input device tool to plan linear surgical trajectories.
- Used **Unity 3D game engine** to simulate environments with different lighting to generate images to **train a CNN model to minimise registration error**, which was utilised for a demo augmented reality app using the Vuforia engine.

Research Intern — Samsung Research Institute, Bangalore, India | 🔍

Dec 2021 – Jun 2022

- Developed a **CNN model 'LiteAR'** for dynamically estimating illumination for mixed reality applications using **spherical harmonics** for A and M series Samsung smartphones.
- Developed an augmented reality app by incorporating LiteAR into the open-source rendering engine Filament.
- Worked on Samsung's proprietary **Neural Radiance Fields (NeRF)** model utilising inverse rendering to generate 3D views from a few 2D images.

PUBLICATIONS

- [1] LiteAR: A Framework to Estimate Lighting for Mixed Reality Sessions for Enhanced Realism | **Advances in Computer Graphics, 2022**
Chinmay Raut, Anamitra Mani, Lakshmi Priya Muraleedharan, Raghavan Velappan | 
- [2] Automated Flexible Needle Trajectory Planning for Keyhole Neurosurgery Using Reinforcement Learning | **IEEE/RSJ International Conference on Intelligent Robots and Systems, 2022**
Jayesh Kumar, Chinmay Raut, Niravkumar Patel | 
- [3] Region of Interest Matters: How Input Region and Feature Extractor Choice Interact for Axillary Lymph Node Metastasis Prediction from Breast Ultrasound | **ICECCME, 2026**
Chinmay Raut, Pradeeba Sridar, Ekta Mukhija

MANUSCRIPTS UNDER REVIEW

- [1] Clinically-Informed Peritumoral Masked Autoencoder for Axillary Lymph Node Status Assessment from Breast Ultrasound | **Under Review, IEEE Journal of Biomedical and Health Informatics**
Chinmay Raut, Pradeeba Sridar, Ekta Mukhija | 
- [2] WAAU-Net: A Wavelet-Domain Adaptation Framework for Population-Invariant Breast Ultrasound Segmentation | **Under Review, Biomedical Signal Processing and Control**
Chinmay Raut, Pradeeba Sridar, Ekta Mukhija | 
- [3] A Decoder-Free Soft-Token Segmenter for Point-of-Care Breast Ultrasound | **Under Review, Journal of Imaging Informatics in Medicine**
Chinmay Raut, Pradeeba Sridar, Ekta Mukhija | 

POSITIONS OF RESPONSIBILITIES & VOLUNTEER WORK

- Teaching Assistant** | *Data Science: Theory & Practice, Introduction to Computation & Visualization* Aug 2022 – May 2023
- Taught Python for data science and C for computation and graphics visualisation in the laboratory sessions for both courses.
 - Evaluated assignments and code submissions for 60+ students.
- Project Management Head** | *Nirmaan, IITM* |  Sep 2020 – Jun 2021
- Led a team of 10 and managed pre-seed funding, legal support, and talent acquisition for 40+ student startups. Conducted project reviews that helped 15+ teams develop an MVP.
- Teaching Resource for Underprivileged Children** | *National Service Scheme, IITM* Sep 2018 – May 2019
- Taught mathematics and science to children from underprivileged communities in Chennai.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, JavaScript, HTML/CSS
Frameworks: PyTorch, PySpark, Flask, Streamlit, LangChain, LangGraph
Developer Tools: Git, Docker

MISCELLANEOUS

Extra-curricular activities

- Filmmaking, Media club, Writing club.
- Part of the institute triathlon team, water polo team.

Optimization Engineer - Avishkar Hyperloop Team

Jan 2019 - May 2020

- Worked on optimizing the power consumption for linear induction motor as a part of the only team from Asia reaching SpaceX Hyperloop Pod Competition finals.